INCH-POUND

MIL-DTL-13777/3B
11 May 2011
SUPERSEDING
MIL-C-13777/3A
1 September 1966

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL 16, 17, 18, 19, AND 20 CONDUCTORS

Inactive for new design after 1 July 2002.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-13777.

REQUIREMENTS:

Dimensions and configurations: See applicable figure and design data for the following cable types. The cable types are specified in table I. Design data information is specified in table II.

TABLE I. Cable types.

| | | | | |
|---|---|---|---|---|
| 160636 | S161375S | 180675 | S190965S | 200935 |
| S160878S | 170752 | 180848 | 191110S | |
| 161065S | 170874 | S190875 | 191140 | |

AMSC N/A

FSC 6145

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TABLE II. Design data.[1/]

| | Design data | | | | |
|---|---|---|---|---|---|
| Type designation | 160636 | | S160878S | | 161065S |
| Figure number | 1 | | 2 | | 3 |
| Total wires | 16 | | 16 | | 16 |
| Number of conductors & AWG # | 16/#20 | 8/#18 | 8/#18S | 12/#18 | 4/#18 |
| Insulation | | | | | |
| Min average thickness | 0.012” | | 0.015” | | 0.015” |
| Spark test voltage | 3000 | | 3000 | | 3000 |
| Inspection test voltage | 1500 | | 1500 | | 1500 |
| Coaxial jacket insulation | | | | | |
| (a) Minimum average thickness | | | | | |
| Cabling | | | | | |
| Layer 1 | Filler | | Filler | | Filler |
| Layer 2 | | | | | |
| (a) Number of wires | 5 | | 6 | | 4 |
| (b) AWG # | #20 | | #18 | | #18S |
| (c) Maximum lay | 2.0” | | 2.50” | | 3.75” |
| Layer 3 | | | | | |
| (a) Number of wires | 11 | 2 | | 8 | 12 |
| (b) AWG # | #20 | #18 | | #18S | #18 |
| (c) Maximum lay | 4.5” | | 5.00” | | 6.00” |
| Sheath | | | | | |
| Number of layers | 2 | | 1 | | 2 |
| Total thickness min. | 0.109” | | 0.080” | | 0.100” |
| Minimum OD cable | 0.616” | | 0.863” | | 1.050” |
| Maximum OD cable | 0.656” | | 0.893” | | 1.080” |
| Maximum weight (lbs./1000ft.) | | | 538 | | |

See notes at end of table.



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TABLE II. Design data – Continued.[1/]

| | Design data | | | | |
|---|---|---|---|---|---|
| Type designation | S161375S | | | 170752[2/] | 170874[3/] |
| Figure number | | 4 | | 5 | 5 |
| Total wires | | 16 | | 17 | 17 |
| Number of conductors & AWG # | 2/#18S | 10/#18S | 4/#6 | 17/#18 | 17/#16 |
| Insulation | | | | | |
| Minimum average thickness | | 0.015” | 0.060” | 0.015” | 0.015” |
| Spark test voltage | | 3000 | 7500 | 3000 | 3000 |
| Inspection test voltage | | 1500 | 2500 | 1500 | 1500 |
| Cabling | | | | | |
| Layer 1 | | | | Filler | Filler |
| (a) Number of wires | | 2 | | | |
| (b) AWG # | | #18[4/] | | | |
| (c) Maximum lay | | 3.00” | | | |
| Layer 2 | | | | | |
| (a) Number of wires | 5 | 5 | 4 | 6 | 6 |
| (b) AWG # | #18[5/] | #18S | #6 | #18 | #16 |
| (c) Maximum lay | | 8.00” | | | |
| Layer 3 | | | | | |
| (a) Number of wires | | | | | |
| (b) AWG # | | | | #18 | #16 |
| (c) Maximum lay | | | | 5.00” | 6.00” |
| Sheath | | | | | |
| Number of layers | | 2 | | 2 | 2 |
| Total thickness minimum | | 0.100” | | 0.110” | 0.109” |
| Minimum OD cable | | 1.360” | | 0.737” | 0.849” |
| Maximum OD cable | | 1.390” | | 0.767” | 0.899” |

See notes at end of table.



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TABLE II. Design data – Continued.[1/]

| | Design data | | | |
|---|---|---|---|---|
| Type designation | 180675 | | 180848 | S190875 [6/] |
| Figure number | 6 | | 7 | 8 |
| Total wires | 18 | | 18 | 19 |
| Number of conductors & AWG # | 18/#20 | 14/#18 | 4/#16 | 19/#18 |
| Insulation | | | | |
| Minimum average thickness | 0.015” | | 0.015” | 0.015” |
| Spark test voltage | 3000 | | 3000 | 3000 |
| Inspection test voltage | 1500 | | 1500 | 1500 |
| Cabling | | | | |
| Layer 1 | Filler | | Filler | Filler |
| Layer 2 | | | | |
| (a) Number of wires | 6 | 2 | 4 | 7 |
| (b) AWG # | #20 | #18 | #16 | #18 |
| (c) Maximum lay | | | | |
| Layer 3 | | | | |
| (a) Number of wires | 12 | | 12 | |
| (b) AWG # | #20 | | #18 | #18 |
| (c) Maximum lay | | | | 5.00” |
| Sheath | | | | |
| Number of layers | 1 | | 2 | 1 |
| Total thickness minimum | 0.080” | | 0.098” | 0.094” |
| Minimum OD cable | 0.650” | | 0.823” | 0.850” |
| Maximum OD cable | 0.700” | | 0.873” | 0.900” |

See notes at end of table.



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TABLE II. Design data – Continued.[1/]

| | Design data | | | | | |
|---|---|---|---|---|---|---|
| Type designation | S190965S | | | | 191110S | |
| Figure number | | 9 | | | 8 | |
| Total wires | | 19 | | | 19 | |
| Number of conductors & AWG # | 8/#18S | | 11/#18 | 7/#20 | | 12/#20S |
| Insulation | | | | | | |
| Minimum average thickness | | 0.015” | | | 0.012” | |
| Spark test voltage | | 3000 | | | 3000 | |
| Inspection test voltage | | 1500 | | | 1500 | |
| Cabling | | | | | | |
| Layer 1 | | Filler | | | Filler | |
| Layer 2 | | | | | | |
| (a) Number of wires | | 8 | | | 7 | |
| (b) AWG # | | #18 | | | #20 | |
| (c) Maximum lay | | 2.50” | | | 4.0” | |
| Layer 3 | | | | | | |
| (a) Number of wires | 8 | | 3 | | 12 | |
| (b) AWG # | #18S | | #18 | | #20S | |
| (c) Maximum lay | | 5.00” | | | 8.0” | |
| Sheath | | | | | | |
| Number of layers | | 1 | | | 2 | |
| Total thickness min | | 0.080” | | | 0.140” | |
| Minimum OD cable | | 0.950” | | | 1.080” | |
| Maximum OD cable | | 0.980” | | | 1.140” | |

See notes at end of table.



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TABLE II. Design data – Continued.[1/]

| | Design data | | | |
|---|---|---|---|---|
| Type designation | 191140 | | 200935 | |
| Figure number. | 8 | | 10 | |
| Total wires | 19 | | 20 | |
| Number of conductors & AWG # | 19/#12 | 13/#18 | | 7/#15 |
| Insulation | | | | |
| Minimum average thickness | 0.020” | | 0.015” | |
| Spark test voltage | 4000 | | 3000 | |
| Inspection test voltage | 2000 | | 1500 | |
| Cabling | | | | |
| Layer 1 | Filler | | Filler | |
| Layer 2 | | | | |
| (a) Number of wires | 7 | | 7 | |
| (b) AWG # | #12 | | #15 | |
| (c) Maximum lay | | | 3.25” | |
| Layer 3 | | | | |
| (a) Number of wires | 12 | | 13 | |
| (b) AWG # | #12 | | #18 | |
| (c) Maximum lay | | | 4.25” | |
| Sheath | | | | |
| Number of layers | 2 | | 2 | |
| Total thickness minimum | 0.098” | | 0.125” | |
| Minimum OD cable | 1.115” | | 0.920” | |
| Maximum OD cable | 1.165” | | 0.950” | |
| Maximum weight (lbs/1000ft.) | | | 540 | |

1/ Jacketed coaxial wires
2/ 11 & 1 filler
3/ 11 W/fillers
4/ Shield applied over twisted pair of AWG #18 – Maximum twist lay 3.00”
5/ Assembly with max lay of 3.00”
6/ 12 W/filler

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.



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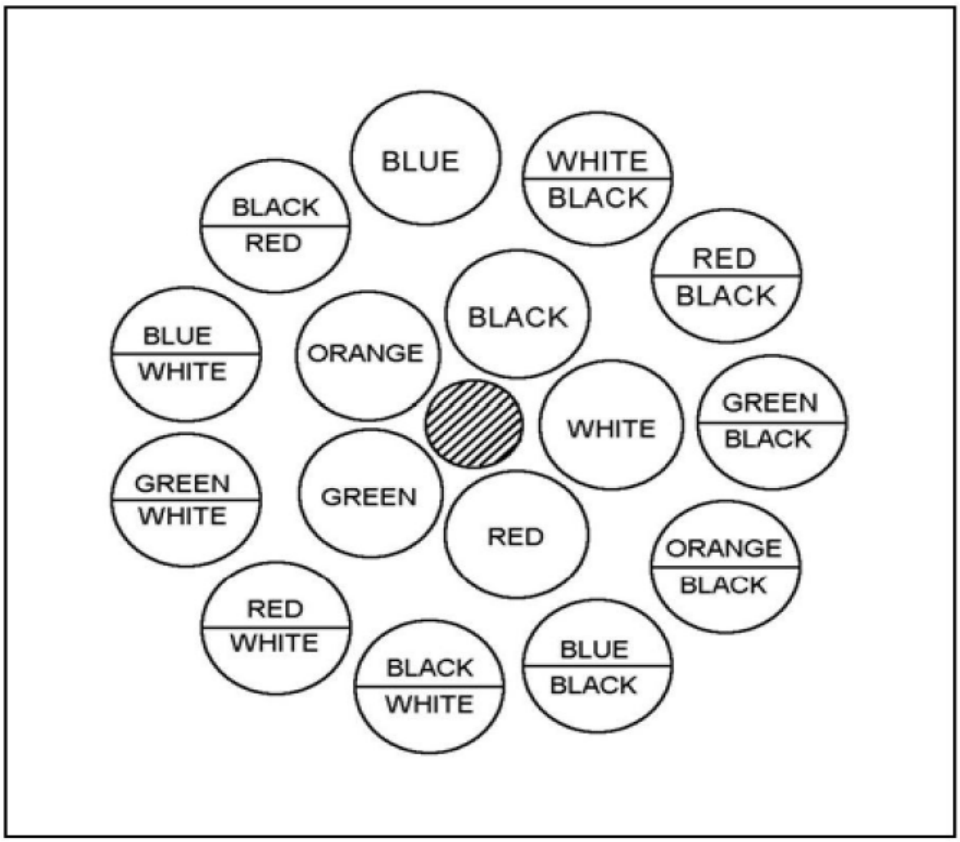



FIGURE 1. Cable type (160636).



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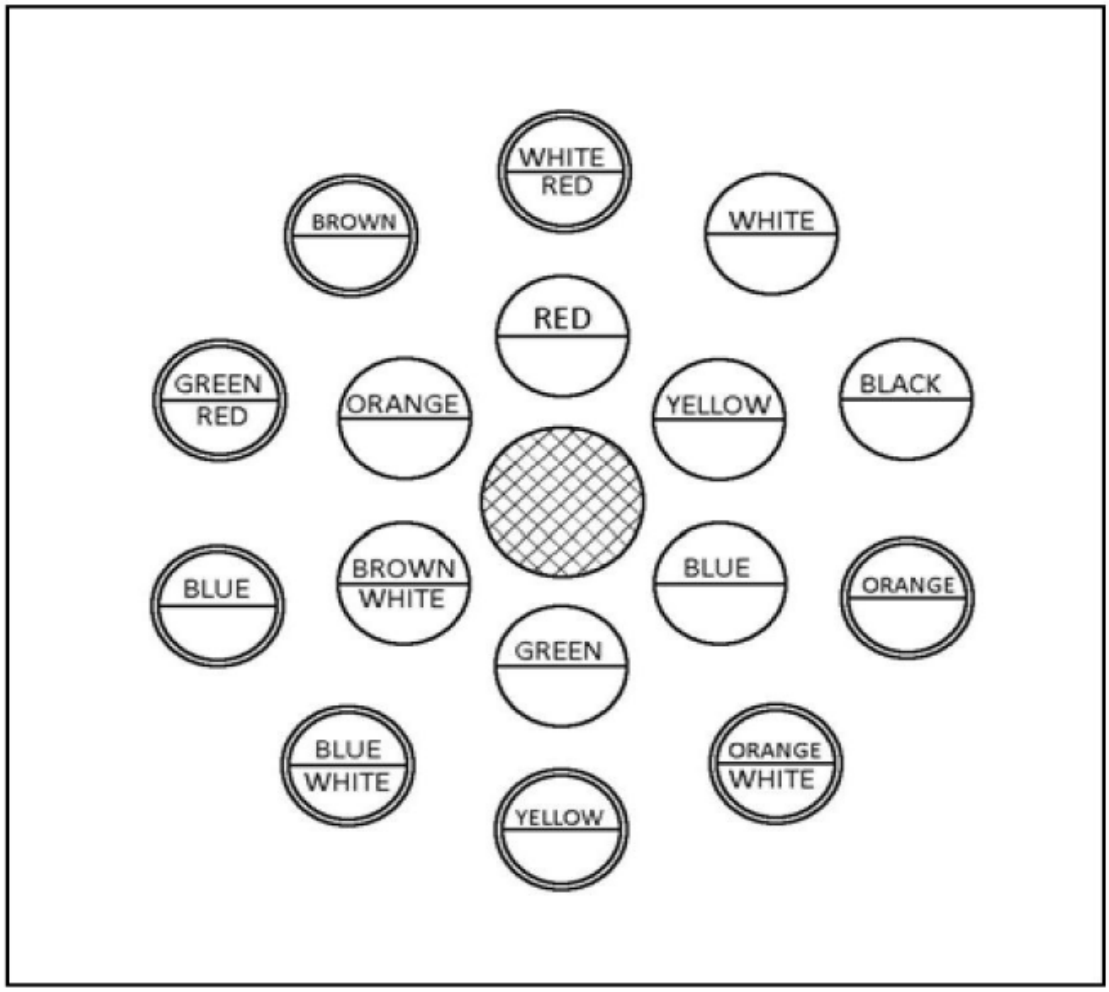



FIGURE 2. Cable type (S160878S).



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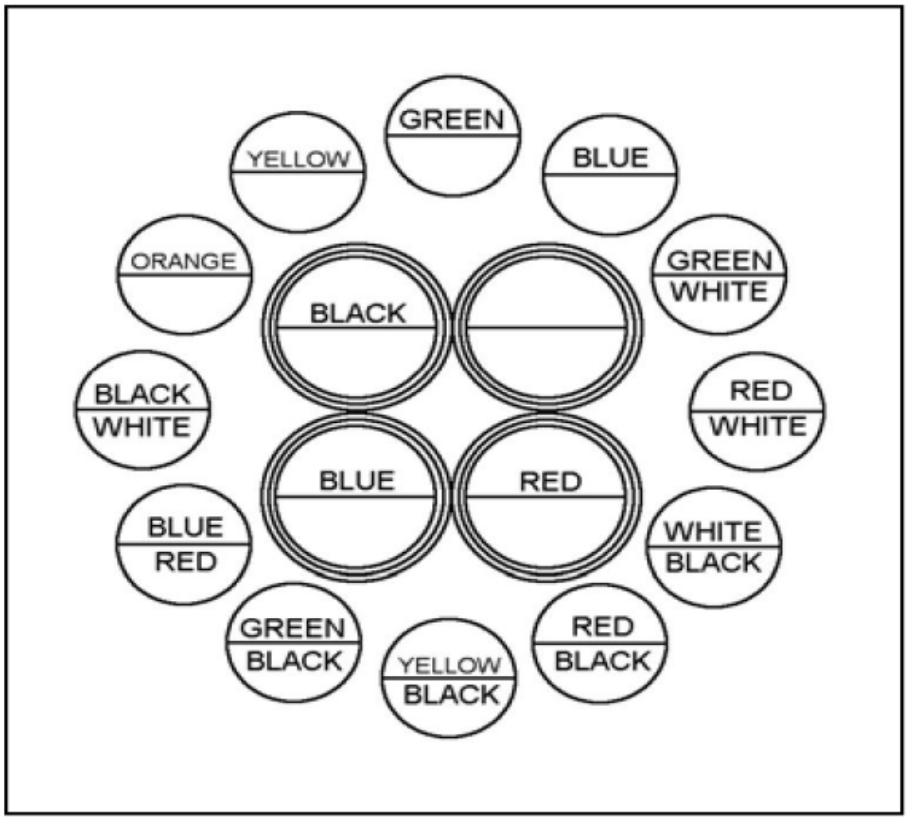



FIGURE 3. Cable type (161065S).



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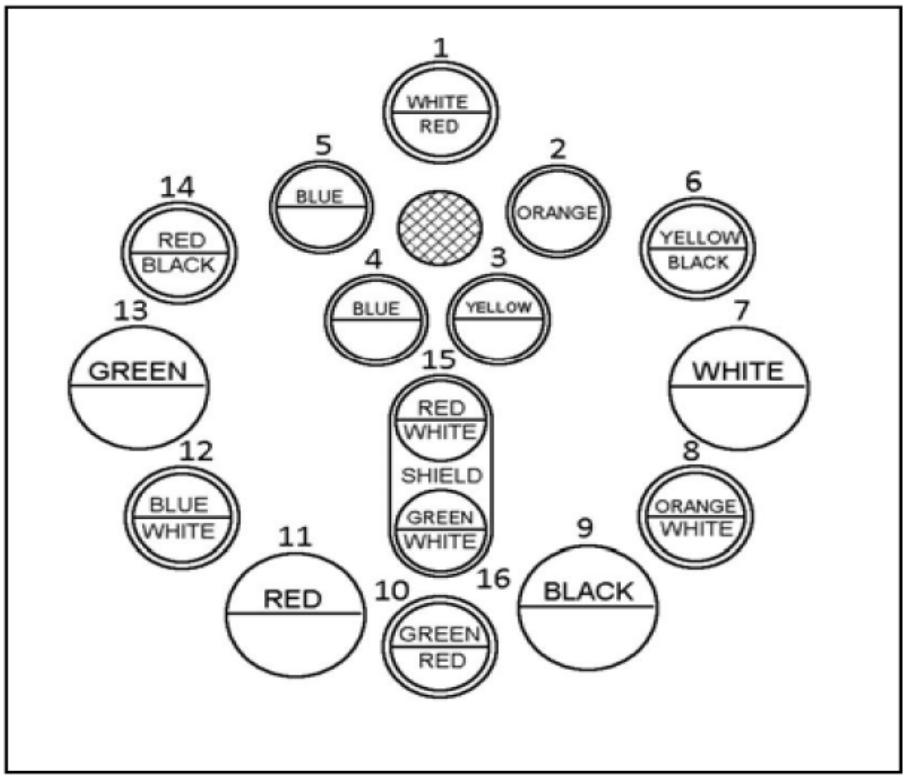



FIGURE 4. Cable type (S161375S).



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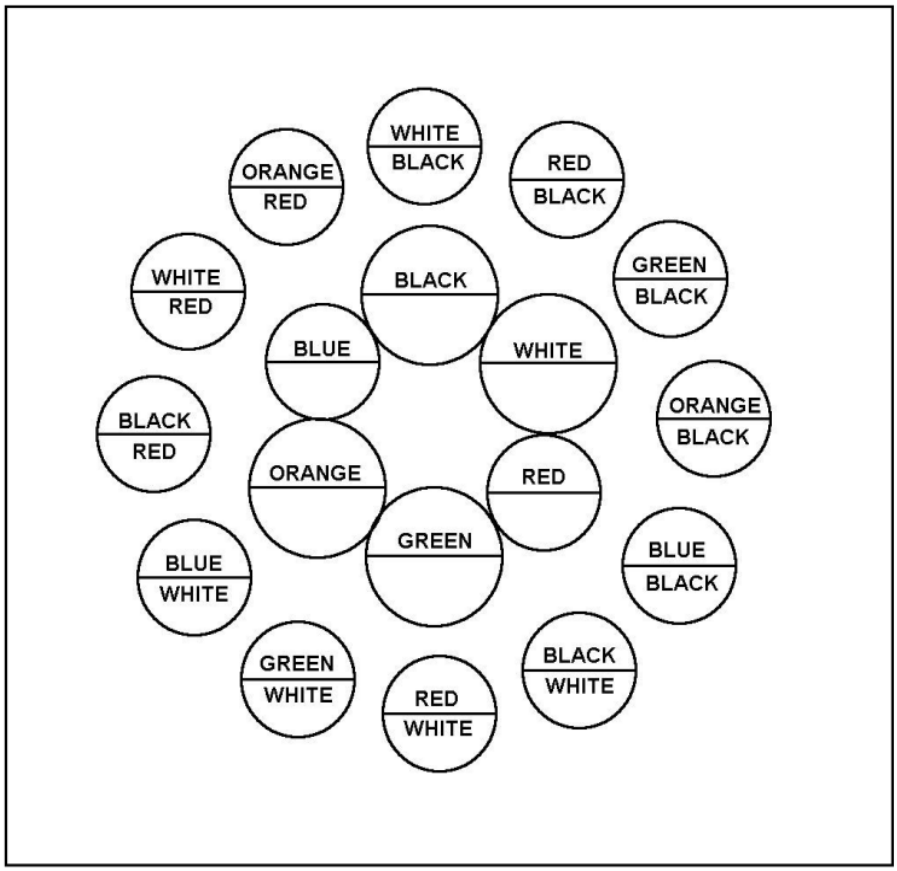



FIGURE 5. Cable types (170752, 170874).



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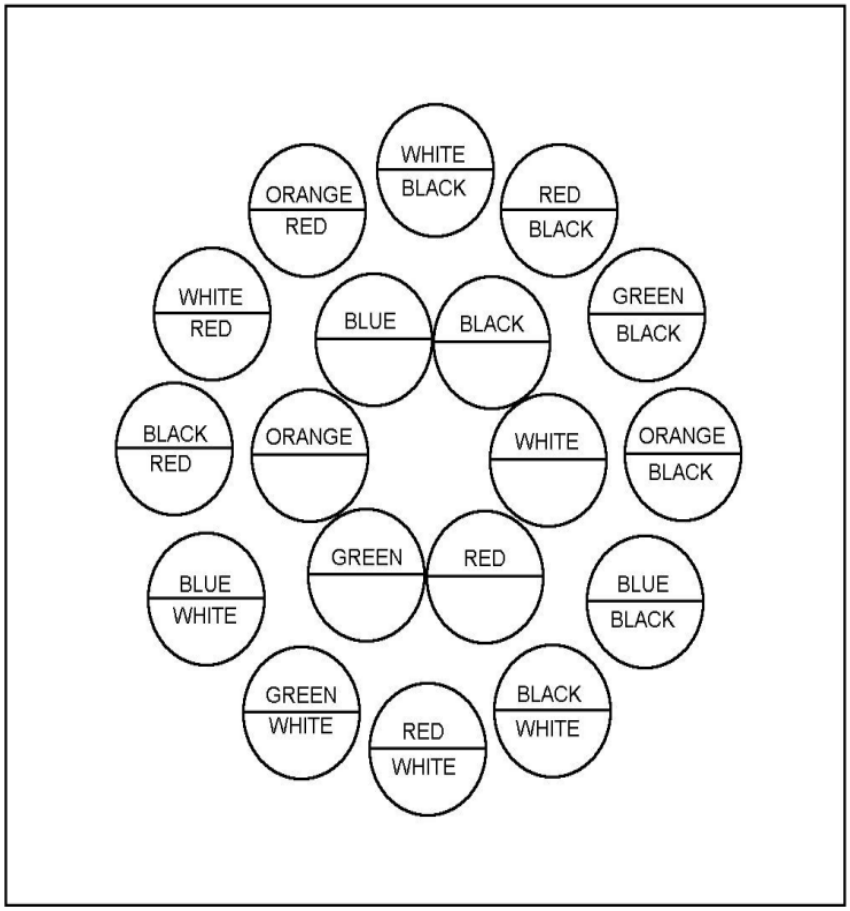



FIGURE 6. Cable type (180675).



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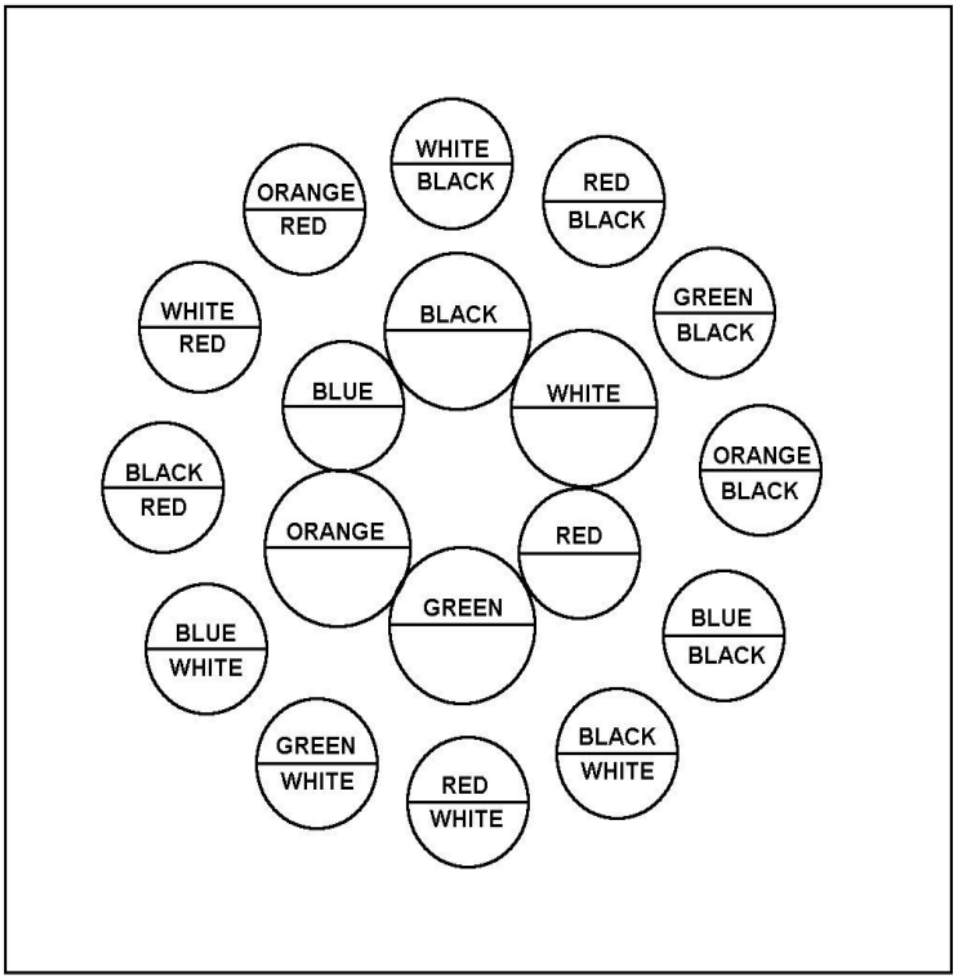
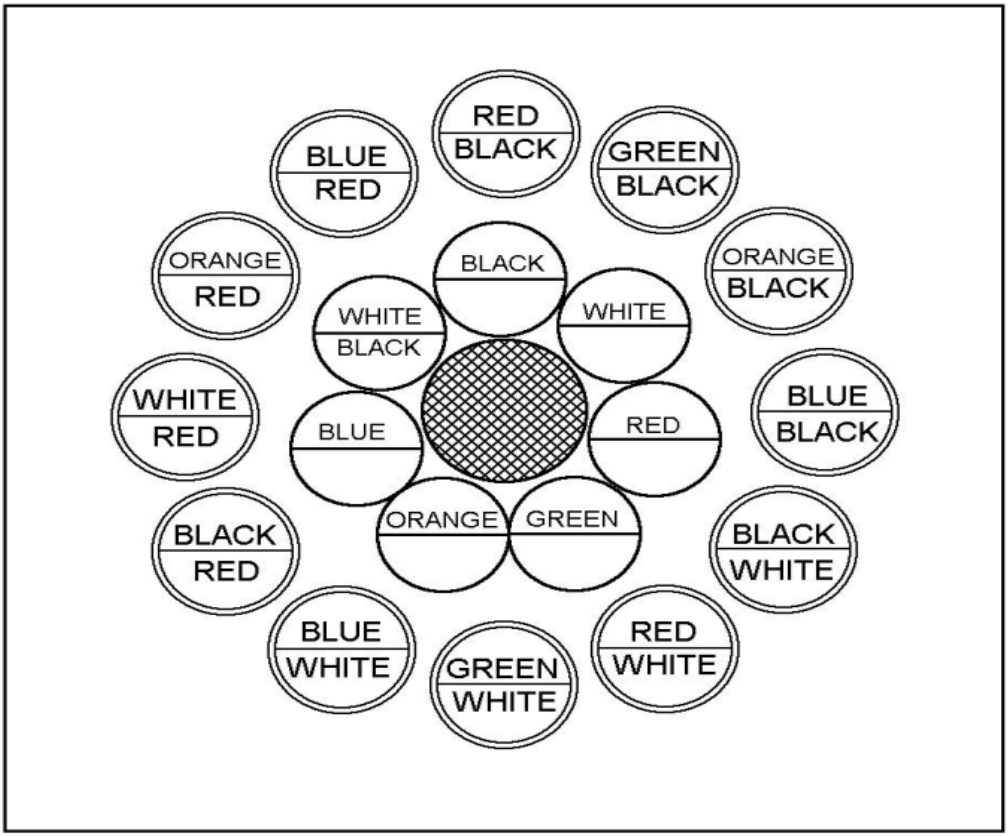



FIGURE 7. Cable type (180848).



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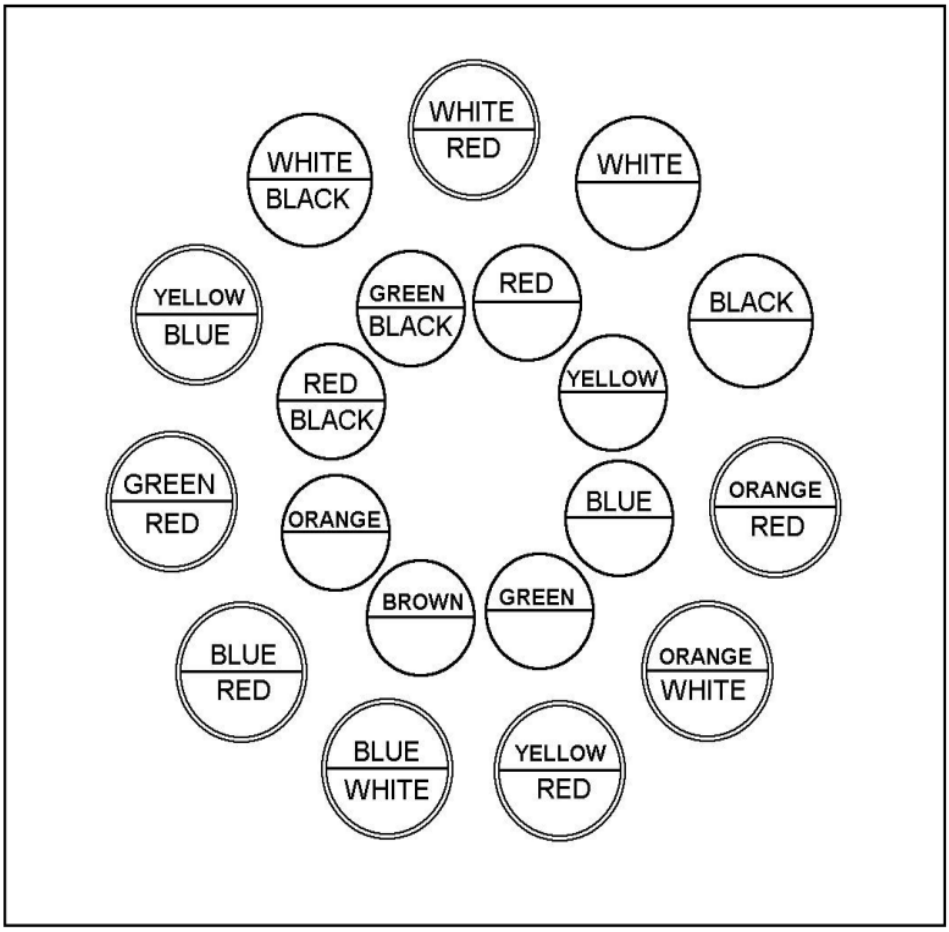



NOTE: *Double circle indicates shielded conductors.

FIGURE 8. Cable types (S190875, 19110S*, 191140).



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NOTE: Double circle indicates shielded conductors.

FIGURE 9. Cable type (S190965S).



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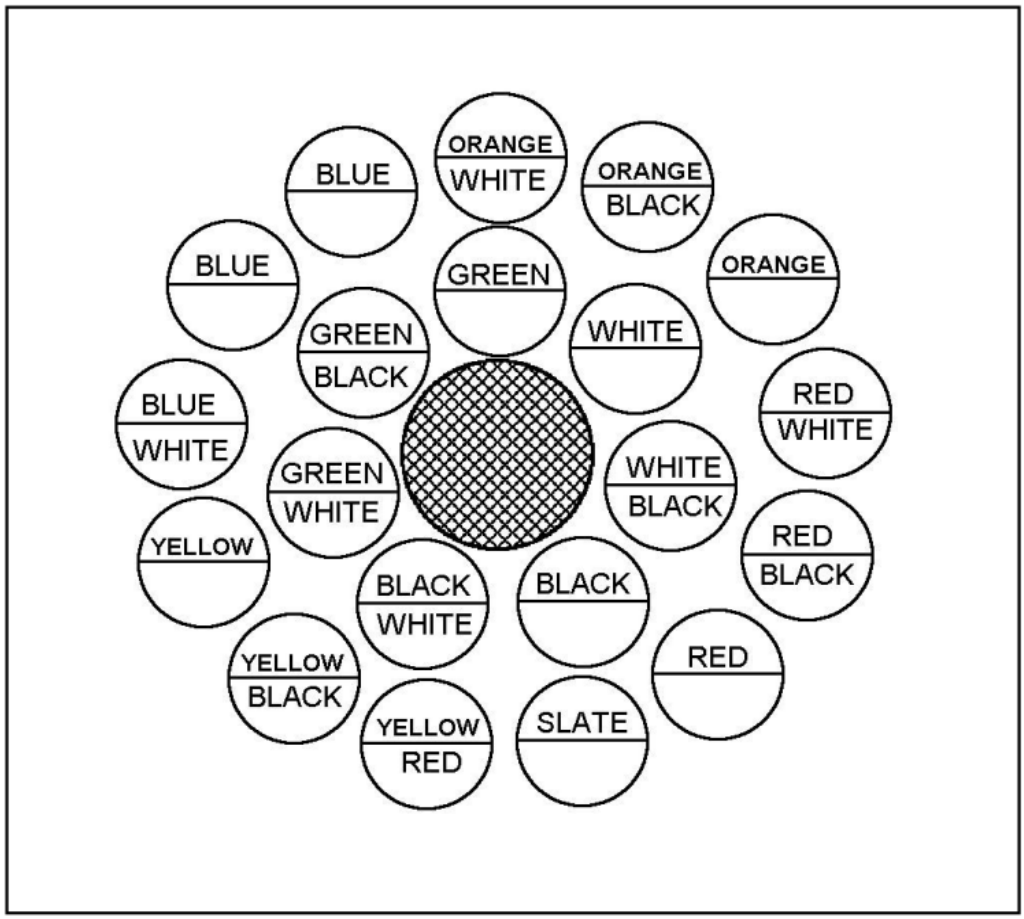



FIGURE 10. Cable type (200935).



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CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 85
DLA CC

Preparing activity:
DLA – CC

(Project 6145-2011-005)

Review activities:
Army – CE, CR, MI
Navy – MC
Air Force – 19

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